6D_1 is the Case where the two curled dimensions form a torus, T^2, the product of two circles. That is, $T^2 = S^1 x S^1$.

Note. T^2 is homogeneous but not isotropic. Thus, the solution is expected to have more free parameters than the 6D_2 case where the compact curled space is a 2-sphere, S^2.

Notation and Setup:

Starting Metric: g_{00}=A[t, r], g_{11}=-a[t]^2 B[r], g_{22}=-r^2 a[t]^2, g_{33}=-r^2 a[t]^2 Sin[θ]^2, g_{44}=-a[t]^2 p_4^2 $\mathcal{D}_4$[r], g_{55}=-a[t]^2 p_5^2 $\mathcal{D}_5$[r],
a(t) not constant

Corresponding Line Element

ds^2 = A[t, r] dt^2 - a[t]^2 B[r] dr^2 - r^2 a[t]^2 $d\theta^2$ - r^2 a[t]^2 Sin[θ]^2 $d\phi^2$ - a[t]^2 p_4^2 $\mathcal{D}_4$[r] $d\psi_4^2$ - a[t]^2 p_5^2 $\mathcal{D}_5$[r] $d\psi_5^2$

Output Display Simplification - Parameters are sometimes suppressed
Prime (if any) means derivative wrt r.
Dot (if any) means derivative wrt t.
Examples: **A = A(t,r)**, **$A' = \partial_r A(t,r)$** and **$\dot{A} = \partial_t A(t,r)$**

Lambda Vacuum Equations: The equations are $R_{\mu\nu} = \Lambda\, g_{\mu\nu}$. They are expressed as $\frac{R_{\mu\nu}}{g_{\mu\nu}} = \Lambda$ when $g_{\mu\nu} \neq 0$, and $R_{\mu\nu} = 0$ when $g_{\mu\nu} = 0$. The $g_{\mu\nu}$ matrix used is a diagonal matrix. Since off-diagonal $R_{\mu\nu}$ equations rarely appear, almost all the equations are expressed by dividing out $g_{\mu\nu}$.

GRW Equations: An exact solution to the Λ vacuum field equations that has (1) free parameter a, (2) C = 1, and (3) parameter $\mathcal{A}$ such that $\mathcal{A}$(t) = c^2 is the Friedmann Continuity Equation (which generates the a(t) formula).

Appendix: Rationale for GRW Equations Assumptions

In order to explain why we made the assumptions in the main body that were used to solve this case, we need to explain what the most general assumption is and why it reduced to the specific assumption used.

To do this, we need to solve this case from scratch, develop the general assumptions, and solve the field equations using them.

The line element above contains only very limited assumptions. It explicitly shows A, B, $\mathcal{D}_4$, and $\mathcal{D}_5$, and it implies C=1.

It shows that $\mathcal{D}_4$ and $\mathcal{D}_5$ are assumed to be functions of r.

They are not required to be functions of r. We simply allow it and see if the field equations reduce them to constants, instead.

In summary, this 6D GRW exploration begins with the assumptions:

a(t) is not constant,

A(t,r) is a function of t and r,

B(r) is a free parameter whose formula we seek,

C(r) = 1,

and $\mathcal{D}_4[r]$ and $\mathcal{D}_5[r]$ are free parameters whose formulas we seek.

Having said this, the resulting field equations are too difficult to solve without making a further assumption. We can solve for A and any two of B, $\mathcal{D}_4$, and $\mathcal{D}_5$ but not all three. This seems to be a classic symptom of "leftover gauge freedom." So, after solving for A, we will discuss assumption options.

The initial Lambda field equations are:

$$R_{00}: \quad \frac{5\,\ddot{a}}{a\,A} - \frac{5\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{A'}{a^2\,A\,r\,B} + \frac{(A')^2}{4\,a^2\,A^2\,B} + \frac{A'\,B'}{4\,a^2\,A\,B^2} - \frac{A'\,\mathcal{D}_4'}{4\,a^2\,A\,B\,\mathcal{D}_4} - \frac{A'\,\mathcal{D}_5'}{4\,a^2\,A\,B\,\mathcal{D}_5} - \frac{A''}{2\,a^2\,A\,B} = \Lambda$$

$$R_{01}: \quad -\frac{2\,\dot{a}\,A'}{a\,A} = 0$$

$$R_{11}: \quad \frac{4\,\dot{a}^2}{a^2\,A} + \frac{\ddot{a}}{a\,A} - \frac{\dot{a}\,\dot{A}}{2\,a\,A^2} + \frac{(A')^2}{4\,a^2\,A^2\,B} + \frac{B'}{a^2\,r\,B^2} + \frac{A'\,B'}{4\,a^2\,A\,B^2} + \frac{B'\,\mathcal{D}_4'}{4\,a^2\,B^2\,\mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{4\,a^2\,B\,\mathcal{D}_4^2} + \frac{B'\,\mathcal{D}_5'}{4\,a^2\,B^2\,\mathcal{D}_5} + \frac{(\mathcal{D}_5')^2}{4\,a^2\,B\,\mathcal{D}_5^2} - \frac{A''}{2\,a^2\,A\,B} - \frac{\mathcal{D}_4''}{2\,a^2\,B\,\mathcal{D}_4} - \frac{\mathcal{D}_5''}{2\,a^2\,B\,\mathcal{D}_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2} - \frac{1}{a^2\,r^2\,B} + \frac{4\,\dot{a}^2}{a^2\,A} + \frac{\ddot{a}}{a\,A} - \frac{\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{A'}{2\,a^2\,A\,r\,B} + \frac{B'}{2\,a^2\,r\,B^2} - \frac{\mathcal{D}_4'}{2\,a^2\,r\,B\,\mathcal{D}_4} - \frac{\mathcal{D}_5'}{2\,a^2\,r\,B\,\mathcal{D}_5} = \Lambda$$

$$R_{33}: \quad \frac{1}{a^2\,r^2} - \frac{1}{a^2\,r^2\,B} + \frac{4\,\dot{a}^2}{a^2\,A} + \frac{\ddot{a}}{a\,A} - \frac{\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{A'}{2\,a^2\,A\,r\,B} + \frac{B'}{2\,a^2\,r\,B^2} - \frac{\mathcal{D}_4'}{2\,a^2\,r\,B\,\mathcal{D}_4} - \frac{\mathcal{D}_5'}{2\,a^2\,r\,B\,\mathcal{D}_5} = \Lambda$$

$$R_{44}: \quad \frac{4\,\dot{a}^2}{a^2\,A} + \frac{\ddot{a}}{a\,A} - \frac{\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{\mathcal{D}_4'}{a^2\,r\,B\,\mathcal{D}_4} - \frac{A'\,\mathcal{D}_4'}{4\,a^2\,A\,B\,\mathcal{D}_4} + \frac{B'\,\mathcal{D}_4'}{4\,a^2\,B^2\,\mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{4\,a^2\,B\,\mathcal{D}_4^2} - \frac{\mathcal{D}_4'\,\mathcal{D}_5'}{4\,a^2\,B\,\mathcal{D}_4\,\mathcal{D}_5} - \frac{\mathcal{D}_4''}{2\,a^2\,B\,\mathcal{D}_4} = \Lambda$$

$$R_{55}: \quad \frac{4\,\dot{a}^2}{a^2\,A} + \frac{\ddot{a}}{a\,A} - \frac{\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{\mathcal{D}_5'}{a^2\,r\,B\,\mathcal{D}_5} - \frac{A'\,\mathcal{D}_5'}{4\,a^2\,A\,B\,\mathcal{D}_5} + \frac{B'\,\mathcal{D}_5'}{4\,a^2\,B^2\,\mathcal{D}_5} - \frac{\mathcal{D}_4'\,\mathcal{D}_5'}{4\,a^2\,B\,\mathcal{D}_4\,\mathcal{D}_5} + \frac{(\mathcal{D}_5')^2}{4\,a^2\,B\,\mathcal{D}_5^2} - \frac{\mathcal{D}_5''}{2\,a^2\,B\,\mathcal{D}_5} = \Lambda$$

R_{33} is redundant, the same as R_{22}.

Since a(t) ≠ constant, from R_{01} we get that

A(t,r) = $\mathcal{A}$(t), a new unknown, strictly a function of t, which again tells us that A(t,r) cannot be a function of r.

R_{01} is eliminated, and the remaining equations become

R_{00}: $$\frac{5\,\ddot{a}}{a\,\mathcal{A}} - \frac{5\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} = \Lambda$$

R_{11}:
$$\frac{4\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{\ddot{a}}{a\,\mathcal{A}} - \frac{\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} + \frac{B'}{a^2\,r\,B^2} + \frac{B'\,D_4'}{4\,a^2\,B^2\,D_4} + \frac{(D_4')^2}{4\,a^2\,B\,D_4^2} + \frac{B'\,D_5'}{4\,a^2\,B^2\,D_5} + \frac{(D_5')^2}{4\,a^2\,B\,D_5^2} - \frac{D_4''}{2\,a^2\,B\,D_4} - \frac{D_5''}{2\,a^2\,B\,D_5} = \Lambda$$

R_{22}: $$\frac{1}{a^2\,r^2} - \frac{1}{a^2\,r^2\,B} + \frac{4\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{\ddot{a}}{a\,\mathcal{A}} - \frac{\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} + \frac{B'}{2\,a^2\,r\,B^2} - \frac{D_4'}{2\,a^2\,r\,B\,D_4} - \frac{D_5'}{2\,a^2\,r\,B\,D_5} = \Lambda$$

R_{44}: $$\frac{4\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{\ddot{a}}{a\,\mathcal{A}} - \frac{\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} - \frac{D_4'}{a^2\,r\,B\,D_4} + \frac{B'\,D_4'}{4\,a^2\,B^2\,D_4} + \frac{(D_4')^2}{4\,a^2\,B\,D_4^2} - \frac{D_4'\,D_5'}{4\,a^2\,B\,D_4\,D_5} - \frac{D_4''}{2\,a^2\,B\,D_4} = \Lambda$$

R_{55}: $$\frac{4\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{\ddot{a}}{a\,\mathcal{A}} - \frac{\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} - \frac{D_5'}{a^2\,r\,B\,D_5} + \frac{B'\,D_5'}{4\,a^2\,B^2\,D_5} - \frac{D_4'\,D_5'}{4\,a^2\,B\,D_4\,D_5} + \frac{(D_5')^2}{4\,a^2\,B\,D_5^2} - \frac{D_5''}{2\,a^2\,B\,D_5} = \Lambda$$

We can immediately find $\mathcal{A}$ from R_{00}:

$$\mathcal{A}(t) = \frac{5\,\dot{a}[t]^2}{-5\,k + \Lambda\,a[t]^2} \quad \text{where curvature } k \in \mathbb{R} \text{ is a constant of integration.}$$

R_{00} is eliminated, and the remaining equations become

R_{11}: $$-\frac{4\,k}{a^2} + \Lambda + \frac{B'}{a^2\,r\,B^2} + \frac{B'\,D_4'}{4\,a^2\,B^2\,D_4} + \frac{(D_4')^2}{4\,a^2\,B\,D_4^2} + \frac{B'\,D_5'}{4\,a^2\,B^2\,D_5} + \frac{(D_5')^2}{4\,a^2\,B\,D_5^2} - \frac{D_4''}{2\,a^2\,B\,D_4} - \frac{D_5''}{2\,a^2\,B\,D_5} = \Lambda$$

R_{22}: $$-\frac{4\,k}{a^2} + \frac{1}{a^2\,r^2} - \frac{1}{a^2\,r^2\,B} + \Lambda + \frac{B'}{2\,a^2\,r\,B^2} - \frac{D_4'}{2\,a^2\,r\,B\,D_4} - \frac{D_5'}{2\,a^2\,r\,B\,D_5} = \Lambda$$

R_{44}: $$-\frac{4\,k}{a^2} + \Lambda - \frac{D_4'}{a^2\,r\,B\,D_4} + \frac{B'\,D_4'}{4\,a^2\,B^2\,D_4} + \frac{(D_4')^2}{4\,a^2\,B\,D_4^2} - \frac{D_4'\,D_5'}{4\,a^2\,B\,D_4\,D_5} - \frac{D_4''}{2\,a^2\,B\,D_4} = \Lambda$$

R_{55}: $$-\frac{4\,k}{a^2} + \Lambda - \frac{D_5'}{a^2\,r\,B\,D_5} + \frac{B'\,D_5'}{4\,a^2\,B^2\,D_5} - \frac{D_4'\,D_5'}{4\,a^2\,B\,D_4\,D_5} + \frac{(D_5')^2}{4\,a^2\,B\,D_5^2} - \frac{D_5''}{2\,a^2\,B\,D_5} = \Lambda$$

This is the point at which an additional assumption is needed in order to solve the equations. Recall that in 5D we found that

$$B = \frac{1}{D_4} = \frac{1}{1 - k\,r^2 + \frac{\chi}{r}}.$$

In 6D, we can solve using D_4 or D_5. Since the circles of radius p_4 and p_5 are initially symmetrical, it does not matter which parameter we solve for first. We will always choose to solve for D_5 first.

There are several assumptions that can extend this to 6D. For example,

$B = \frac{1}{\mathcal{D}_5}$, which eventually leads to $\mathcal{D}_5 = \frac{1}{1 - k r^2 + \frac{\chi}{r}}$ (with k=0).

This can also be expressed as

$$B = \frac{1}{\mathcal{D}_4 \mathcal{D}_5} \text{ where } \mathcal{D}_4 = 1$$

which suggests other assumptions, like

$$B = \frac{1}{\mathcal{D}_4 \mathcal{D}_5} \text{ where } \mathcal{D}_4 = \mathcal{D}_5 = \sqrt{1 - k r^2 + \frac{\chi}{r}}$$

or $$B = \frac{1}{\mathcal{D}_4 \mathcal{D}_5} \text{ where } \mathcal{D}_4 = \left(1 - k r^2 + \frac{\chi}{r}\right)^{\alpha} \text{ and } \mathcal{D}_5 = \left(1 - k r^2 + \frac{\chi}{r}\right)^{1-\alpha}$$

In addition, we don't yet know the 6D formula for B. We currently have

$$\text{4D:} \quad B = \frac{1}{1 - k r^2}$$

$$\text{5D:} \quad B = \frac{1}{1 - k r^2 + \frac{\chi}{r}}$$

The most general assumption thus appears to be:

$$B = \frac{1}{\mathcal{D}_4 \mathcal{D}_5}.$$

where $\mathcal{D}_4$ and $\mathcal{D}_5$ are free parameters whose formulas we seek.

$\mathcal{D}_4$ and $\mathcal{D}_5$ can be any of the formulas list above, or indeed, any formulas.

With this assumption, the remaining equations now become

$$R_{11}: \quad -\frac{4k}{a^2} + \Lambda - \frac{\mathcal{D}_5 \mathcal{D}_4'}{a^2 r} - \frac{\mathcal{D}_4 \mathcal{D}_5'}{a^2 r} - \frac{\mathcal{D}_4' \mathcal{D}_5'}{2a^2} - \frac{\mathcal{D}_5 \mathcal{D}_4''}{2a^2} - \frac{\mathcal{D}_4 \mathcal{D}_5''}{2a^2} = \Lambda$$

$$R_{22}: \quad -\frac{4k}{a^2} + \frac{1}{a^2 r^2} + \Lambda - \frac{\mathcal{D}_4 \mathcal{D}_5}{a^2 r^2} - \frac{\mathcal{D}_5 \mathcal{D}_4'}{a^2 r} - \frac{\mathcal{D}_4 \mathcal{D}_5'}{a^2 r} = \Lambda$$

$$R_{44}: \quad -\frac{4k}{a^2} + \Lambda - \frac{\mathcal{D}_5 \mathcal{D}_4'}{a^2 r} - \frac{\mathcal{D}_4' \mathcal{D}_5'}{2a^2} - \frac{\mathcal{D}_5 \mathcal{D}_4''}{2a^2} = \Lambda$$

$$R_{55}: \quad -\frac{4k}{a^2} + \Lambda - \frac{\mathcal{D}_4 \mathcal{D}_5'}{a^2 r} - \frac{\mathcal{D}_4' \mathcal{D}_5'}{2a^2} - \frac{\mathcal{D}_4 \mathcal{D}_5''}{2a^2} = \Lambda$$

$R_{11}-R_{44}-R_{55}$ can be solved:

$$R_{11}-R_{44}-R_{55}: \quad \left(\frac{4k}{a^2} - \Lambda + \frac{\mathcal{D}_4' \mathcal{D}_5'}{2a^2}\right) = -\Lambda$$

or $$\mathcal{D}_4' \mathcal{D}_5' = -8k$$

Solving this produces two branches:

- **Branch 1:** One of $\mathbb{D}_4$, $\mathbb{D}_5$ is constant. We will choose $\mathbb{D}_4$ to be constant.

  Thus $k = 0$, $B = \frac{1}{\mathbb{D}_5}$, and $\mathbb{D}_4 = 1$.

  This branch is the initial assumption used in the main body.

- **Branch 2:** Neither $\mathbb{D}_4$ nor $\mathbb{D}_5$ is constant. We show that this branch yields no valid solution.

  $$\mathbb{D}_5(r) = \zeta + \int_1^r -\frac{8k}{\mathbb{D}_4'[\rho]}\, d\rho \quad \text{where } \zeta \text{ is a constant of integration.}$$

With the formula for $\mathbb{D}_5$, we can consider R_{11} to be redundant, leaving three equations in the single unknown, $\mathbb{D}_4$:

$$R_{22}: \quad -\frac{4k}{a^2} + \frac{1}{a^2 r^2} + \Lambda - \frac{\zeta \mathbb{D}_4}{a^2 r^2} + \frac{8k \mathbb{D}_4}{a^2 r \mathbb{D}_4'} - \frac{\zeta \mathbb{D}_4'}{a^2 r} - \frac{\mathbb{D}_4 \int_1^r -\frac{8k}{\mathbb{D}_4'}\, d\rho}{a^2 r^2} - \frac{\mathbb{D}_4' \int_1^r -\frac{8k}{\mathbb{D}_4'}\, d\rho}{a^2 r} = \Lambda$$

$$R_{44}: \quad \Lambda - \frac{\zeta \mathbb{D}_4'}{a^2 r} - \frac{\zeta \mathbb{D}_4''}{2a^2} - \frac{\mathbb{D}_4' \int_1^r -\frac{8k}{\mathbb{D}_4'}\, d\rho}{a^2 r} - \frac{\mathbb{D}_4'' \int_1^r -\frac{8k}{\mathbb{D}_4'}\, d\rho}{2a^2} = \Lambda$$

$$R_{55}: \quad \Lambda + \frac{8k \mathbb{D}_4}{a^2 r \mathbb{D}_4'} - \frac{4k \mathbb{D}_4 \mathbb{D}_4''}{a^2 (\mathbb{D}_4')^2} = \Lambda$$

R_{55} can be solved for $\mathbb{D}_4$:

$$\mathbb{D}_4(r) = \Xi + \frac{r^3 \chi}{3} \quad \text{where } \chi \text{ and } \Xi \text{ are constants of integration.}$$

$$\Rightarrow \quad \mathbb{D}_5(r) = \zeta - \frac{8k(-1+r)}{r\chi}$$

$$\Rightarrow \quad B(r) = \frac{1}{\left(\zeta - \frac{8k(-1+r)}{r\chi}\right)\left(\Xi + \frac{r^3\chi}{3}\right)}$$

The remaining two equation are:

$$R_{22}: \quad -\frac{12k}{a^2} + \frac{1}{a^2 r^2} + \frac{32kr}{3a^2} + \Lambda - \frac{\zeta \Xi}{a^2 r^2} + \frac{8k\Xi}{a^2 r^2 \chi} - \frac{4r\zeta\chi}{3a^2} = \Lambda$$

$$R_{44}: \quad -\frac{16k}{a^2} + \frac{16kr}{a^2} + \Lambda - \frac{2r\zeta\chi}{a^2} = \Lambda$$

$$R_{44} \Rightarrow \quad -16k + r(16k - 2\zeta\chi) = 0$$

$$\Rightarrow \quad k = 0 \quad \text{and} \quad \zeta\chi = 0$$

Since $\chi \neq 0$, we get $\zeta = 0$

$$\Rightarrow \quad \mathbb{D}_4(r) = \Xi + \frac{r^3 \chi}{3}$$

$$\Rightarrow \quad \mathbb{D}_5(r) = 0 \text{ not allowed}$$

This concludes this appendix. We have
shown that Branch 2 does not lead to a valid solution.